

Model bake-off — M.I.R.A. agent tasks

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Three representative tasks from the M.I.R.A. agent (planner tool-calling, news-sentiment classification, structured-JSON synthesis) executed against each model through OpenRouter. Measures correctness, latency, tokens, cost.

Summary

Model	Pass	p50 latency	Avg latency	Total tokens	Total cost
GPT-5.4 (openai/gpt-5.4)	3/3	2.8s	3.02s	1,071	\$0.00969
Grok 4.3 (x-ai/grok-4.3)	3/3	6.21s	5.66s	2,914	\$0.00605
DeepSeek V4 Pro (deepseek/deepseek-v4-pro)	3/3	16.42s	17.41s	1,827	\$0.00123

Per-task detail

planner_tool_call

Model	Pass	Latency	Prompt tok	Completion tok	Cost	Note
GPT-5.4	✓	1.77s	202	84	\$0.00177	tools_called=3: get_market_data,get_news_sentiment,get_correlation
Grok 4.3	✓	3.48s	443	399	\$0.00155	tools_called=3: get_market_data,get_news_sentiment,get_correlation
DeepSeek V4 Pro	✓	4.04s	510	174	\$0.00037	tools_called=3: get_market_data,get_news_sentiment,get_correlation

news_sentiment

Model	Pass	Latency	Prompt tok	Completion tok	Cost	Note
GPT-5.4	✓	2.8s	126	194	\$0.00323	classified 5/5
Grok 4.3	✓	6.21s	244	541	\$0.00166	classified 5/5
DeepSeek V4 Pro	✓	31.76s	117	408	\$0.00041	classified 5/5

synthesis_json

Model	Pass	Latency	Prompt tok	Completion tok	Cost	Note
GPT-5.4	✓	4.49s	182	283	\$0.00470	schema valid
Grok 4.3	✓	7.3s	301	986	\$0.00284	schema valid
DeepSeek V4 Pro	✓	16.42s	187	431	\$0.00046	schema valid

Verdict

- **Highest pass rate:** GPT-5.4 (3/3)
- **Fastest (p50):** GPT-5.4 (2.8s)
- **Cheapest:** DeepSeek V4 Pro (\$0.00123)

Methodology

- Driver: `backend/eval/run_model_benchmark.py`.
- Tasks mirror the real M.I.R.A. agent: (1) planner emits tool calls, (2) news-sentiment classifier returns JSON array, (3) synthesizer emits the final `AnalysisReport`-shaped JSON.
- Pass = task-specific structural check (correct tool names called; valid JSON of the right shape; sentiment in `[-1,1]`; exactly 3 `key_findings`).
- Cost = $(\text{prompt} \times \text{in_rate} + \text{completion} \times \text{out_rate}) / 1e6$, rates verified against OpenRouter / `models` on 2026-05-16.

Re-run

```
cd backend && python -m eval.run_model_benchmark
```